

IPv6 (Part-1)

- Topics covered:
 - Hexadecimal (review)
 - Why IPv6?
 - Basics of IPv6
 - Configuring IPv6 addresses

RECAP

What we do till now?

In this video we will cover IPv6.

Up to this point in the course, only IPv4 has been covered.

IPv6 is the future and is starting to take over networks all over the world.

IPv6 brings multiple improvements over IPv4, but there is one main reason for the switch.

IPv6 appears in multiple areas of the official exam topics list.

1.8 Configure and verify IPv6 addressing and prefix

1.9 Compare IPv6 address types

1.9.a Global unicast

1.9.b Unique local

1.9.c Link local

1.9.d Anycast

1.9.e Multicast

1.9.f Modified EUI 64

3.3 Configure and verify IPv4 and IPv6 static routing

3.3.a Default route

3.3.b Network route

3.3.c Host route

3.3.d Floating static

What About IPv5?

Before talking about IPv6, a review of IPv5.

Internet Stream Protocol was developed in the late 1970s.
It was never introduced for public use.

- It was never called IPv5
- It used a value of 5 in the Version field of the IP header

The IPv4 header was covered earlier in the course.

The first field in the IPv4 header is the Version field.

- IPv4 → Version value 4
- Internet Stream Protocol → Version value 5

To avoid confusion, the successor to IPv4 was named IPv6, and it uses a value of 6 in the Version field.

Hexadecimal

Number Notation?

The three numbering systems required for CCNA are:

Binary (Base 2)

- Prefix: 0b
- Digits available: 0, 1
- All numbers are represented using only these two digits

Example ambiguity without prefix:

10

Could mean:

- Decimal 10
- Binary 10 (decimal 2)
- Hexadecimal 10 (decimal 16)

Using 0b10 removes ambiguity.

Decimal (Base 10)

- Prefix: 0d
- Digits available: 0 to 9
- Total of 10 digits

Hexadecimal (Base 16)

- Prefix: 0x
- Digits available:
 - 0–9, A, B, C, D, E, F

Decimal, Binary, Hexadecimal Comparison Chart:

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9

10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Notes from the chart:

- Decimal and hexadecimal are the same from 0–9
- Decimal runs out of digits at 9 and uses two digits for 10
- Hexadecimal uses a **single digit** (A) for decimal 10
- Each hexadecimal digit represents **4 bits**
- Leading zeros are shown to emphasize 4-bit structure

Maximum example:

Binary: 1111

Hex: F

Binary to Hexadecimal Conversion Examples

Let see the Conversion Binary to Hexadecimal there are some steps you can follow and get the answer.

Example 1: 0b11011011 = 0x??

0b11011011

↓ split

1101 1011 -> in Binary

↓ ↓

13 11 -> in Decimal

↓ ↓

D B -> in Hexadecimal

Result: 0b11011011 = 0xDB

Example 2: 0b00101111 = 0x??

0b00101111

↓ split

0010 1111

↓ ↓

2 15

↓ ↓

2 F

Result: 0b00101111 = 0x2F

Example 3: 0b10000001 = 0x??

0b10000001

↓ split

1000 0001

↓ ↓

8 1

↓ ↓

8 1 -> in Hexadecimal

Result: 0b10000001 = 0x81

Hexadecimal to Binary Conversion

Let see the Conversion Binary to Hexadecimal there are some steps you can follow and get the answer.

Example 1: 0xEC = 0b??

0xEC

↓ ↓

E C -> in Hexadecimal

↓ ↓

14 12 -> in Decimal

↓ ↓

1110 1100 -> in Binary

Result: 0xEC = 0b11101100 -> Final Answer

Example 2: 0x2B = 0b??

0x2B

↓ ↓

2 B

↓ ↓

2 11

↓ ↓

0010 1011

Result: 0x2B = 0b00101011

Example 3: 0xD7 = 0b??

0xD7

↓ ↓

D 7

↓ ↓

13 7

↓ ↓

1101 0111

Result: 0xD7 = 0b11010111

Why IPv6?

Why we use IPv6

The main reason for IPv6 is IPv4 address exhaustion.

IPv4 Address Space

- IPv4 address length: 32 bits
- Total IPv4 addresses: 4,294,967,296

This is not enough for the modern Internet.

When IPv4 was designed, the Internet was much smaller, and 32 bits were thought to be sufficient.

IPv4 Address Preservation Techniques

Used as short-term solutions:

- VLSM (Variable-Length Subnet Masks)
- Private IPv4 addresses
- NAT (Network Address Translation)

These help preserve IPv4 but are not long-term solutions.

Long-Term Solution

The long-term solution is to transition to IPv6.

IPv4 Address Assignment Authority

IPv4 addresses are controlled by IANA (Internet Assigned Numbers Authority).

Flow of assignment:

IANA

↓

RIRs (Regional Internet Registries)

↓

ISPs

↓ Customers

Regional Internet Registries (RIRs)

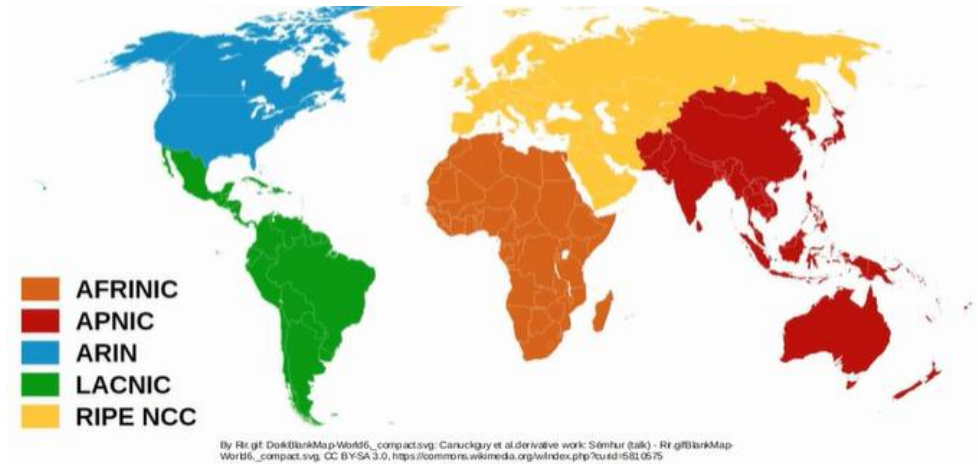
[AFRINIC] → Africa

[APNIC] → Asia-Pacific

[ARIN] → USA, Canada, parts of Caribbean

[LACNIC] → Latin America & Caribbean

[RIPE NCC] → Europe, Middle East, Central Asia



IPv4 Exhaustion Status

- ARIN declared IPv4 exhaustion in September 2015
- LACNIC made its final IPv4 allocation in August 2020
- Other RIRs face similar problems
- New IPv4 addresses are only available if reclaimed

Basics of IPv6

IPv6 — Addressing, Shortening, and Prefix Calculation

Let Understand who ipv6 works and how we use it.

IPv6 Address Size and Address Space

An IPv6 address is 128 bits.

That is 4 times the number of bits in an IPv4 address, which is 32 bits.

At first, it may seem that 4 times the number of bits would mean 4 times the number of addresses, That is incorrect.

Every additional bit doubles the number of possible addresses.

- 32 bits → about 4 billion addresses
- 33 bits → about 8 billion addresses
- 34 bits → about 16 billion addresses

Total Number of IPv6 Addresses

There are: 340,282,366,920,938,463,463,374,607,431,768,211,456 IPv6 addresses.

For comparison, IPv4 has: 4,294,967,296 addresses.

You do not need to memorize the IPv6 number.

IPv6 Address Representation

IPv6 Address in Binary:

```
0010000000000000100001101101111000010110010001011111101010101111010110
010101100010000101111101010110010010010110101100110111101
```

This is a very large number of bits.

Same Address Written As IPv4 (Dotted Decimal):

```
32.1.13.184.89.23.234.189.101.98.23.234.201.45.89.189
```

IPv6 addresses are not written this way.

IPv6 Written in Hexadecimal: 2001 : 0DB8 : 5917 : EABD : 6562 : 17EA : C92D : 59BD /64

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

IPv6 Address Structure

- IPv6 address length: 128 bits
- Each hexadecimal digit = 4 bits
- $128 \div 4 = 32$ hexadecimal characters
- Written as 8 quartets (groups of 4 hex digits)
- Quartets are separated by colons (:)

This is longer than IPv4, but unavoidable because there is 4 times the information

IPv6 Prefix Notation

IPv6 uses slash notation to indicate prefix length.

Example: /64

- First half → network portion
- Second half → host portion

No dotted decimal subnet masks are used.

IPv6 Address Shortening Methods

There are two methods to shorten IPv6 addresses.

Method 1: Removing Leading Zeros

Leading zeros are zeros at the beginning of any quartet.

Leading 0s can be removed
2001:0DB8:000A:001B:20A1:0020:0080:34BD



2001:DB8:A:1B:20A1:20:80:34BD

The zeros still exist logically, but do not need to be written.

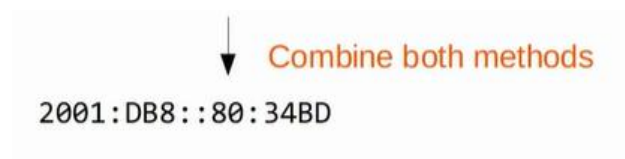
Method 2: Double Colon (::)

Consecutive quartets of all 0s can be replaced with a double colon (::)
2001:0DB8:0000:0000:0000:0000:0080:34BD



2001:0DB8::0080:34BD

Combining Both Methods



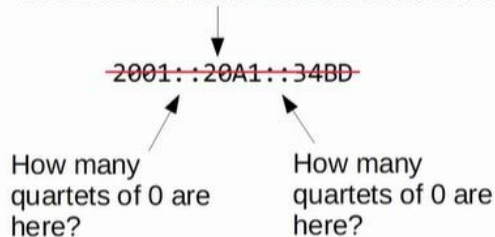
Limitation of Double Colon

Consecutive quarters of all zeros can be abbreviated only once.

Problem Example:

- Consecutive quarters of 0s can only be abbreviated once in an IPv6 address.

2001:0000:0000:0000:20A1:0000:0000:34BD



Correct Shortening:

2001::20A1:0:0:34BD

- Left side: double colon replaces 3 quarters
- Right side: leading zeros removed only

Practice: Shortening IPv6 Addresses

Full IPv6 Address	Shortened IPv6 Address
2000:AB78:0020:01BF:ED89:0000:0000:0001	2000:AB78:20:1BF:ED89::1
FE80:0000:0000:0000:0002:0000:0000:FBE8	FE80::2:0:0:FBE8
AE89:2100:01AC:00F0:0000:0000:0000:020F	AE89:2100:1AC:F0::20F
2001:0DB8:8B00:1000:0002:0BC0:0D07:0099	2001:DB8:8B00:1000:2:BC0:D07:99
2001:0DB8:0000:0000:0000:0000:0000:1000	2001:DB8:::1000

Expanding a Shortened IPv6 Address

Let See how we expand a ipv6 address if they provide in short from.

Step 1: Add Leading Zeros

All quartets must have 4 hexadecimal digits.

FE80::2:0:0:FBE8

Put leading 0s where needed (all quartets should have 4 hexadecimal characters)

FE80: :2:0:0: FBE8



FE80: :0002:0000:0000: FBE8

Step 2: Expand Double Colon

If a double colon is used, replace it with all-0 quartets. Make sure there are 8 quartets in total.

FE80: :0002:0000:0000: FBE8 5 quartets (8 quartets, but only 5 are written)



FE80: :0000:0000:0000:0002:0000:0000: FBE8 8 quartets

Practice: Expanding IPv6 Addresses

Full IPv6 Address	Shortened IPv6 Address
FE80:0000:0000:0000:1010:02FC:0000:0009	FE80::1010:2FC:0:9
2001:0DB8:0001:0B23:2309:0000:0000:00C1	2001:DB8:1:B23:2309::C1
FD00:0000:0000:0000:1000:0689:9000:0CDF	FD00::1000:689:9000:CDF
FF02:0000:0000:0000:0000:0000:0000:0002	FF02::2
0000:0000:0000:0000:0000:0000:0000:0001	::1

IPv6 Network Prefix Basics (Global Unicast Addresses)

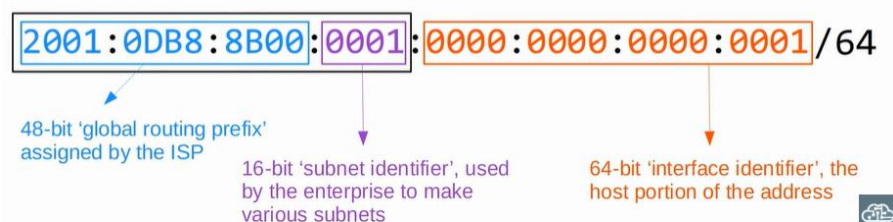
To find an IPv6 prefix:

- Change all host bits to 0
- Prefix length determines where network bits end

Typical IPv6 Allocation

- ISP assigns a **/48**
- Enterprises typically use **/64**
- That means an enterprise has 16 bit to use to make subnet: **16 bits** for subnetting
- Remaining **64 bits** for hosts

IPv6 Address Components (/64 Example)



- First 48 bits: Global routing prefix
- Next 16 bits: Subnet identifier
- Last 64 bits: Host bits

Finding Prefix with /64

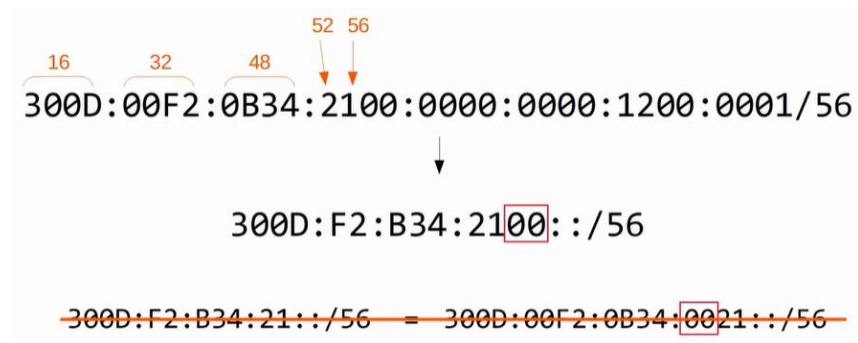
Change host portion to zeros:

2001:0DB8:8B00:0001:0000:0000:0000:0001 /64

↓

2001:DB8:8B00:1:: /64

Prefix Length Multiple of 4 (/56 Example)



Prefix Length Not Multiple of 4 (/93 Example)



Finding the IPv6 prefix

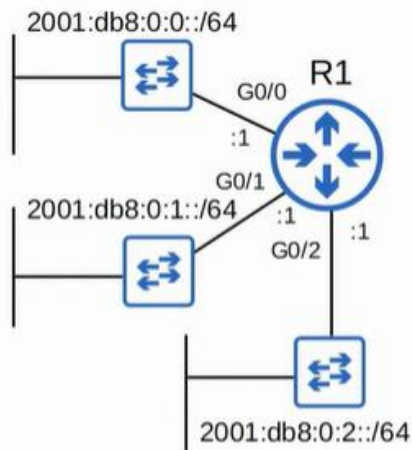
Host Address	Prefix
FE80:0000:0000:0000:4c2c:e2ed:6a89:2a27/9	FE80::/9
2001:0DB8:0001:0B23:BA89:0020:0000:00C1/64	2001:DB8:1:B23::/64
2001:0DB8:0BAD:CAFE:1300:0689:9000:0CDF/71	2001:DB8:BAD:CAFE:1200::/71
2001:0DB8:0000:FEED:0DAD:018F:6001:0DA3/62	2001:DB8:0:FEEC::/62
2001:0DB8:9BAD:BABE:0DE8:AB78:2301:0010/63	2001:DB8:9BAD:BABE::/63

Configuring IPv6 addresses

Transition to IPv6 Configuration

This section shows how to configure IPv6 addresses on router interfaces.

Network Topology Overview



Addressing Explanation

- G0/0 → 2001:db8:0:0::/64
- G0/1 → 2001:db8:0:1::/64
- G0/2 → 2001:db8:0:2::/64

The company was assigned a /48 block.

The last quartet of the prefix is used to create different subnets.

Reserved IPv6 Documentation Address Range

The 2001:db8 range is reserved for examples and documentation.

- It should never be used in real networks.
- It is safe to use in examples and labs.

IPv6 Configuration on Cisco Router

CLI Configuration:

```
R1(config)#
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8:0:0::1/64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/1
R1(config-if)#ipv6 address 2001:db8:0:1::1/64
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int g0/2
R1(config-if)#ipv6 address 2001:0db8:0000:0002:0000:0000:0000:0001/64
R1(config-if)#no shutdown
R1(config-if)#
```

Explanation of Configuration Steps

1] IPv6 Unicast Routing:

ipv6 unicast-routing

- Allows the router to perform IPv6 routing
- Without it, IPv6 packets are not forwarded

2] IPv6 Address Configuration

ipv6 address <address>/<prefix-length>

- Same structure as IPv4 commands
- Uses ipv6 instead of ip
- Shortened, full, or partially shortened addresses are accepted

Remember to use:

no shutdown

to enable the interface.

Verifying IPv6 Configuration

CLI Output:

```
R1#show ipv6 interface brief
GigabitEthernet0/0    [up/up]
    FE80::EF8:22FF:FE36:8500
    2001:DB8::1
GigabitEthernet0/1    [up/up]
    FE80::EF8:22FF:FE36:8501
    2001:DB8:0:1::1
GigabitEthernet0/2    [up/up]
    FE80::EF8:22FF:FE36:8502
    2001:DB8:0:2::1
GigabitEthernet0/3    [administratively down/down]
    unassigned
R1#
```

Verification Notes

- Shortened IPv6 addresses are displayed automatically
- Addresses may appear more shortened than what was typed
- Quartets that were typed to emphasize the network portion can also be included in the double colon

Link-Local Addresses

Each interface shows two IPv6 addresses:

- One manually configured address
- One link-local address

Link-local addresses:

- Are automatically configured when IPv6 is enabled
- Exist even though only one address was configured
- Will be covered in Day 32

IPv4 also has link-local addresses, but they are not automatically enabled.

Video Review Summary

Today What we covered

1. Review of hexadecimal and conversions with binary
2. Why IPv6 is necessary
3. Basics of IPv6 addressing
 - 128-bit addresses
 - Hexadecimal notation
4. Basic IPv6 router configuration
 - Enabling IPv6 routing
 - Assigning IPv6 addresses to interfaces

There is much more to cover about IPv6, but this video provides a solid starting point.